

Ikarians Health Data Science Placement Project

Health Data Mapping, Insight Discovery and AI Readiness

MSc Health Data Science Placement (10 Weeks)

Students: Nyog John Smart, Abdulrahman Alazhari

Project Purpose

To help Ikarians understand how wearable data, onboarding information and patient-reported data can be transformed into meaningful insights that improve patient engagement, clinician decision-making and AI-generated outputs.

The work should bridge health data science, digital health and real-world product development. Students are encouraged to challenge assumptions, identify opportunities and propose practical recommendations throughout the project.

Week 1: Familiarisation and Discovery

Goal

Develop an understanding of Ikarians, the MVP platform, pilot objectives and the available health data ecosystem.

Activities

- Review pitch deck and supporting materials
- Review MVP videos and demonstrations
- Meet founders and technical team
- Understand pilot objectives
- Review examples of available datasets
- Explore current AI and dashboard outputs

Deliverables

- Initial project understanding document (2–3 pages)
- Summary of observations, questions and opportunities

Weeks 2–3: Health Data Mapping

Goal

Understand and document the health data ecosystem supporting Ikarians.

Activities

- Explore wearable datasets
- Explore onboarding assessments
- Explore survey datasets

- Review AI outputs
- Understand how information flows through the platform

Deliverables

- Health Data Ecosystem Map
- Data Source Catalogue
- Initial Data Dictionary
- Data Flow Diagram

Weeks 4–5: Data Quality and Value Assessment

Goal

Evaluate the quality and usefulness of the available health data.

Activities

- Assess completeness and consistency
- Identify underused variables
- Assess opportunities for standardisation
- Assess missing information

Deliverables

- Data Quality Assessment Report
- Signal vs Noise Analysis
- Data Gap Assessment
- Recommendations for future data collection

Weeks 6–7: AI Readiness and Data Optimisation

Goal

Explore how health data should be structured before being consumed by AI systems.

Activities

- Investigate data aggregation approaches
- Explore methods for summarising health information
- Review opportunities to improve AI context quality

Deliverables

- AI Readiness Framework
- Proposed Health Summary Structure
- Recommendations for data prioritisation
- Suggested approach for AI-ready inputs

Week 8: Insight Discovery

Goal

Identify meaningful insights that could be generated from the available data.

Activities

- Explore patterns and trends
- Investigate adherence indicators
- Investigate engagement indicators
- Explore behaviour change opportunities

Deliverables

- Insight Opportunity Report
- Patient insight recommendations
- Professional insight recommendations
- Recommendations for future exploration

Week 9: Product Recommendations

Goal

Translate findings into practical recommendations for Ikarians.

Activities

- Review current outputs and dashboards
- Identify opportunities for simplification
- Prioritise future development opportunities

Deliverables

- Product Recommendations Report
- Dashboard Recommendations
- MVP Improvement Suggestions
- Future Health Data Strategy Recommendations

Week 10: Final Reporting and Presentation

Goal

Consolidate findings into a practical roadmap for Ikarians.

Activities

- Finalise reports
- Prepare presentation materials
- Reflect on project outcomes
- Develop recommendations for next steps

Deliverables

- Executive Summary
- Final Placement Report
- Final Presentation
- Recommendations Roadmap

Success Criteria

Success will be measured by whether the recommendations are practical, evidence-informed and valuable for an early-stage digital health company.

Independent Thinking

Students are encouraged to challenge assumptions and propose new ideas throughout the placement.

Guiding Principle

The purpose of this placement is not to build production systems. The aim is to help Ikarians transform complex, multi-source health data into meaningful, structured and actionable data for insights.